Lecture 14: Convex KKT Conditions and Support Vector Machines

MA3236 Non-Linear Programming

Hu Yukuan

yukuan.hu@nus.edu.sg
S17-08-21

Quick review: KKT SONC and SOSC

$C_{\text{crit}}(\mathbf{x}^*)$: critical cone at $\mathbf{x}^*$; $\quad \mathbf{H}_L = \nabla^2_{\mathbf{xx}} L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$.

Suppose the data are C^2, $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ is a KKT triple, and $\mathcal{S}$ is the feasible set.

Theorem 14.1

If $\mathbf{x}^*$ is a local minimizer and LICQ holds at $\mathbf{x}^*$, then

$$\boxed{\mathbf{d}^T \mathbf{H}_L \mathbf{d} \geq 0 \qquad \forall \mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*)}.$$

Theorem 14.2

If it holds that

$$\boxed{\mathbf{d}^T \mathbf{H}_L \mathbf{d} > 0 \qquad \forall \mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*) \setminus \{\mathbf{0}\}},$$

then there exist $c, \varepsilon > 0$ such that

$$f(\mathbf{x}) \geq f(\mathbf{x}^*) + c\|\mathbf{x} - \mathbf{x}^*\|^2 \quad (\mathbf{x} \in \mathcal{S},\ \|\mathbf{x} - \mathbf{x}^*\| < \varepsilon).$$

Hence $\mathbf{x}^*$ is a strict local minimizer.

When $C_{\text{crit}}(\mathbf{x}^*)$ is a linear subspace

Let the columns of $\mathbf{Z}$ form a basis of $C_{\text{crit}}(\mathbf{x}^*)$, so $C_{\text{crit}}(\mathbf{x}^*) = \text{Range}(\mathbf{Z})$.

$$\mathbf{d}^T \mathbf{H}_L \mathbf{d} \geq 0 \quad \forall \mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*) \iff \mathbf{Z}^T \mathbf{H}_L \mathbf{Z} \succeq 0.$$

$\mathbf{Z}^T \mathbf{H}_L \mathbf{Z}$ is also called the reduced Hessian.

When $C_{\text{crit}}(\mathbf{x}^*)$ is a linear subspace

Let the columns of $\mathbf{Z}$ form a basis of $C_{\text{crit}}(\mathbf{x}^*)$, so $C_{\text{crit}}(\mathbf{x}^*) = \text{Range}(\mathbf{Z})$.

$$\mathbf{d}^T \mathbf{H}_L \mathbf{d} \geq 0 \quad \forall \mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*) \iff \mathbf{Z}^T \mathbf{H}_L \mathbf{Z} \succeq 0.$$

$\mathbf{Z}^T \mathbf{H}_L \mathbf{Z}$ is also called the reduced Hessian.

$J := \{j : h_j(\mathbf{x}^*) = 0\}$, $J_+ := \{j \in J : \nu_j^* > 0\}$, $J_0 := J \setminus J_+$.

$$C_{\text{crit}}(\mathbf{x}^*) = \left\{ \mathbf{d} : \begin{array}{ll} \nabla g_i(\mathbf{x}^*)^T \mathbf{d} = 0, & i = 1{:}m, \\ \nabla h_j(\mathbf{x}^*)^T \mathbf{d} = 0, & j \in J_+, \\ \nabla h_j(\mathbf{x}^*)^T \mathbf{d} \leq 0, & j \in J_0. \end{array} \right\}.$$

1. **Equality constraints only:** $C_{\text{crit}}(\mathbf{x}^*) = \{\mathbf{d} : \nabla g_i(\mathbf{x}^*)^T \mathbf{d} = 0\}$.
2. **Strict complementarity:** $h_j(\mathbf{x}^*) = 0 \Rightarrow \nu_j^* > 0$, so $J_0 = \varnothing$ and

$$C_{\text{crit}}(\mathbf{x}^*) = \{\mathbf{d} : \nabla g_i^T \mathbf{d} = 0,\ \nabla h_j^T \mathbf{d} = 0\ (j \in J)\}.$$

Reduced-Hessian example

$$\min_{\mathbf{x}\in\mathbf{R}^3} \; f(\mathbf{x}) = x_1^2 - x_2^2 + x_3^2 \qquad \text{s. t.} \qquad g(\mathbf{x}) = x_1 = 0.$$

LICQ holds at $\mathbf{x}^* = \mathbf{0}$,

$$\text{KKT FONC}: \quad \nabla f(\mathbf{0}) = \mathbf{0}, \qquad \nabla g(\mathbf{0}) = (1,0,0)^{\mathsf{T}}, \qquad \lambda^* = 0.$$

$$\mathbf{H}_L = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad \mathbf{Z} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{Z}^{\mathsf{T}}\mathbf{H}_L\mathbf{Z} = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix} \not\succeq 0.$$

KKT SONC implies that $\mathbf{x}^* = \mathbf{0}$ cannot be a local minimizer.

When $C_{\text{crit}}(\mathbf{x}^*)$ is only a cone

$$\min\ f(\mathbf{x}) = \frac{1}{4}(x_1^2 + x_2^2) + x_1 x_2 \quad \text{s. t.} \quad h_1(\mathbf{x}) = -x_1 \leq 0,\ h_2(\mathbf{x}) = -x_2 \leq 0.$$

At $\mathbf{0}$, KKT FONC: $\nabla f(\mathbf{0}) = \mathbf{0}$, $\nabla h_1(\mathbf{0}) = (-1, 0)^{\mathsf{T}}$, $\nabla h_2(\mathbf{0}) = (0, -1)^{\mathsf{T}}$, and $\boldsymbol{\nu}^* = \mathbf{0}$. Moreover,

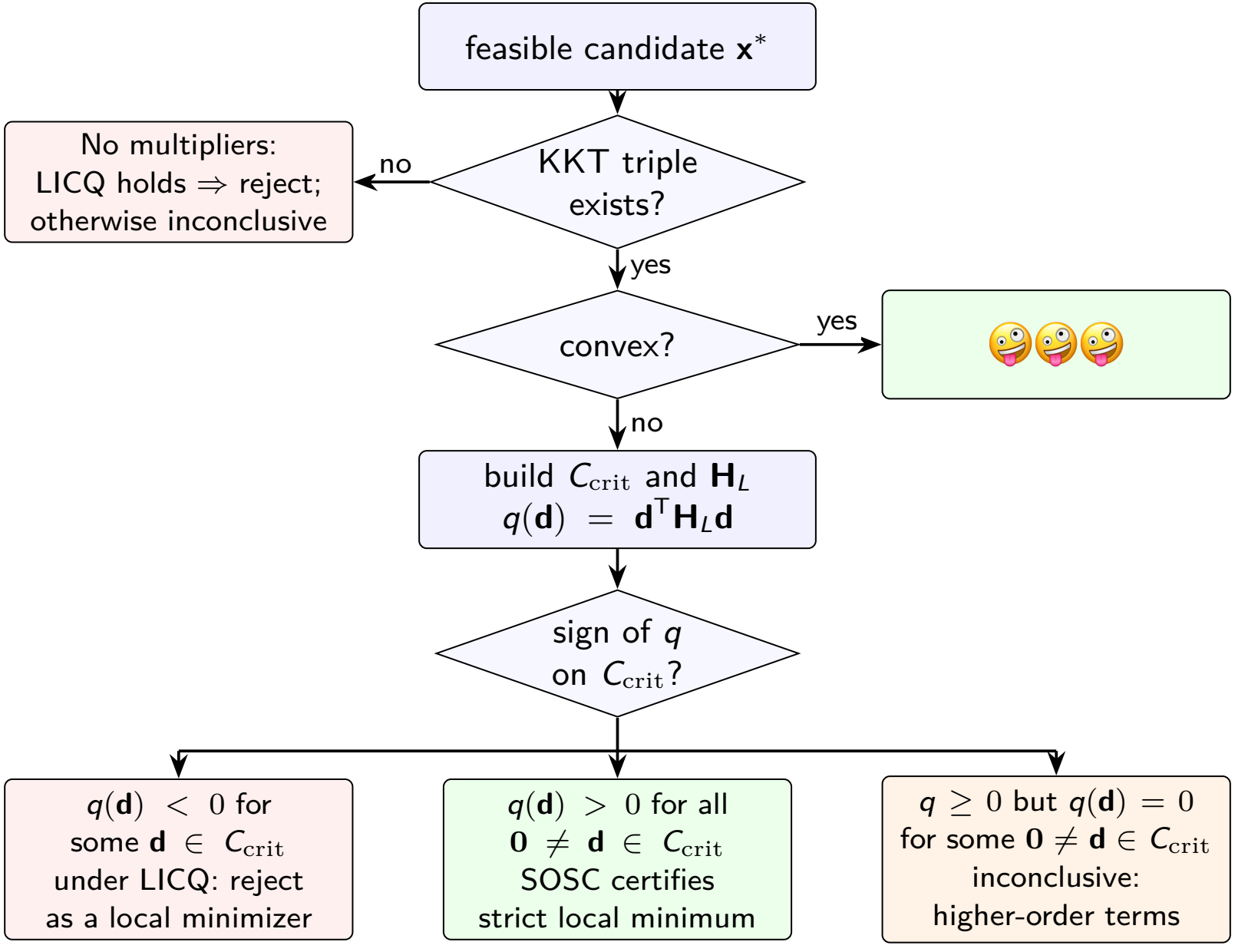
$$C_{\text{crit}}(\mathbf{0}) = \mathbb{R}^2_+, \quad \mathbf{H}_L = \begin{pmatrix} 1/2 & 1 \\ 1 & 1/2 \end{pmatrix} \quad \text{has eigenvalues } \frac{3}{2} \text{ and } -\frac{1}{2},$$

but for $\mathbf{0} \neq \mathbf{d} \in \mathbb{R}^2_+$,

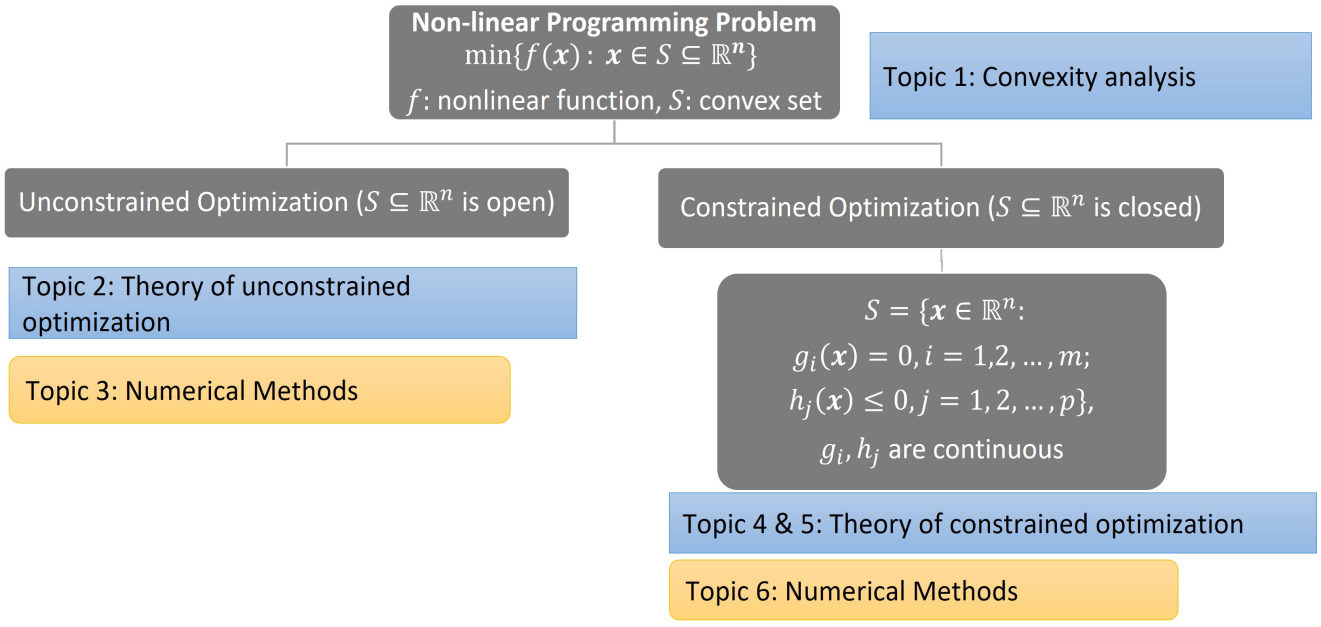
$$\mathbf{d}^{\mathsf{T}} \mathbf{H}_L \mathbf{d} = \frac{1}{2}(d_1^2 + d_2^2) + 2 d_1 d_2 > 0.$$

KKT SOSC succeeds because only the feasible cone matters.

The incomplete KKT first-order and second-order workflow



Topics



In Topic 4, we study optimality conditions for **constrained optimization**.

- KKT first-order necessary conditions and constraint qualifications
- KKT second-order necessary and sufficient conditions
- KKT first-order necessary and sufficient conditions for convex optimization and SVM

Recap: convexity gives an exact test

Let f be smooth and convex on an open convex $\mathcal{D}$, $\mathcal{C} \subseteq \mathcal{D}$ be convex.

Unconstrained convex optimization (L04)

If $\mathcal{C}$ is open, then $\nabla f(\mathbf{x}^*) = \mathbf{0} \iff \mathbf{x}^*$ is a global minimizer.

Recap: convexity gives an exact test

Let f be smooth and convex on an open convex $\mathcal{D}$, $\mathcal{C} \subseteq \mathcal{D}$ be convex.

Unconstrained convex optimization (L04)

If $\mathcal{C}$ is open, then $\nabla f(\mathbf{x}^*) = \mathbf{0} \iff \mathbf{x}^*$ is a global minimizer.

Abstract general convex optimization (L04)

Given $\mathbf{x}^* \in \mathcal{C}$,

$$\mathbf{x}^* \in \arg\min_{\mathbf{x} \in \mathcal{C}} f(\mathbf{x}) \quad \iff \quad \nabla f(\mathbf{x}^*)^\mathsf{T}(\mathbf{x} - \mathbf{x}^*) \geq 0 \quad \forall \mathbf{x} \in \mathcal{C}.$$

The latter one is a variational inequality (VI) on $\mathcal{C}$.

VI is abstract and uses every feasible point. Any explicit algebraic form?

The canonical convex optimization

$$\min_{\mathbf{x}\in\mathcal{C}} f(\mathbf{x}), \qquad \mathcal{C} := \left\{ \mathbf{x} \in \mathcal{D} : \begin{array}{ll} g_i(\mathbf{x}) = \mathbf{a}_i^\mathsf{T}\mathbf{x} - b_i = 0, & i = 1, \ldots, m, \\ h_j(\mathbf{x}) \leq 0, & j = 1, \ldots, p \end{array} \right\}.$$

- $\mathcal{D}$ is open and convex.
- f and all h_j are smooth convex functions on $\mathcal{D}$.
- All equalities $g_i = 0$ are affine.

NB. Affine equalities suffice for a convex feasible set; they are not necessary for every representation.

For the canonical convex optimization, a computable alternative to VI?

Expected answer: KKT FONC (with LICQ)!

VI + LICQ $\Rightarrow$ KKT FONC

Suppose that $\mathbf{x}^* \in \mathcal{C}$ satisfies LICQ and

$$\nabla f(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) \geq 0, \quad \forall\, \mathbf{x} \in \mathcal{C}.$$

Under LICQ, KKT FONC $\Leftrightarrow \nabla f(\mathbf{x}^*)^{\mathsf{T}}\mathbf{d} \geq 0$ for any $\mathbf{d} \in T_{\mathcal{C}}(\mathbf{x}^*)$.
Farkas' alternative: exactly one of the following holds.

- $\nabla f(\mathbf{x}^*) + \nabla \mathbf{g}(\mathbf{x}^*)\boldsymbol{\lambda}^* + \nabla \mathbf{h}_J(\mathbf{x}^*)\boldsymbol{\nu}_J^* = \mathbf{0}$ for some $\boldsymbol{\lambda}^* \in \mathbf{R}^m$ and $\boldsymbol{\nu}_J^* \in \mathbf{R}_+^{|J|}$;
- there is $\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*)$ such that $\nabla f(\mathbf{x}^*)^{\mathsf{T}}\mathbf{d} < 0$.

VI + LICQ $\Rightarrow$ KKT FONC

Suppose that $\mathbf{x}^* \in \mathcal{C}$ satisfies LICQ and

$$\nabla f(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) \geq 0, \quad \forall\, \mathbf{x} \in \mathcal{C}.$$

Pick any $\mathbf{d} \in T_{\mathcal{C}}(\mathbf{x}^*) = T_{\text{lin}}(\mathbf{x}^*)$ (due to LICQ) and the corresponding

$$\{\mathbf{x}^{(k)} \in \mathcal{C}\} \to \mathbf{x}^*, \qquad \{t^{(k)} > 0\} \to 0, \qquad \frac{\mathbf{x}^{(k)} - \mathbf{x}^*}{t^{(k)}} \to \mathbf{d}.$$

Under LICQ, KKT FONC $\Leftrightarrow \nabla f(\mathbf{x}^*)^{\mathsf{T}}\mathbf{d} \geq 0$ for any $\mathbf{d} \in T_{\mathcal{C}}(\mathbf{x}^*)$.
Farkas' alternative: exactly one of the following holds.

- $\nabla f(\mathbf{x}^*) + \nabla \mathbf{g}(\mathbf{x}^*)\boldsymbol{\lambda}^* + \nabla \mathbf{h}_J(\mathbf{x}^*)\boldsymbol{\nu}_J^* = \mathbf{0}$ for some $\boldsymbol{\lambda}^* \in \mathbf{R}^m$ and $\boldsymbol{\nu}_J^* \in \mathbf{R}_+^{|J|}$;
- there is $\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*)$ such that $\nabla f(\mathbf{x}^*)^{\mathsf{T}}\mathbf{d} < 0$.

VI + LICQ $\Rightarrow$ KKT FONC

Suppose that $\mathbf{x}^* \in \mathcal{C}$ satisfies LICQ and

$$\nabla f(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) \geq 0, \quad \forall\, \mathbf{x} \in \mathcal{C}.$$

Pick any $\mathbf{d} \in T_{\mathcal{C}}(\mathbf{x}^*) = T_{\text{lin}}(\mathbf{x}^*)$ (due to LICQ) and the corresponding

$$\{\mathbf{x}^{(k)} \in \mathcal{C}\} \to \mathbf{x}^*, \qquad \{t^{(k)} > 0\} \to 0, \qquad \frac{\mathbf{x}^{(k)} - \mathbf{x}^*}{t^{(k)}} \to \mathbf{d}.$$

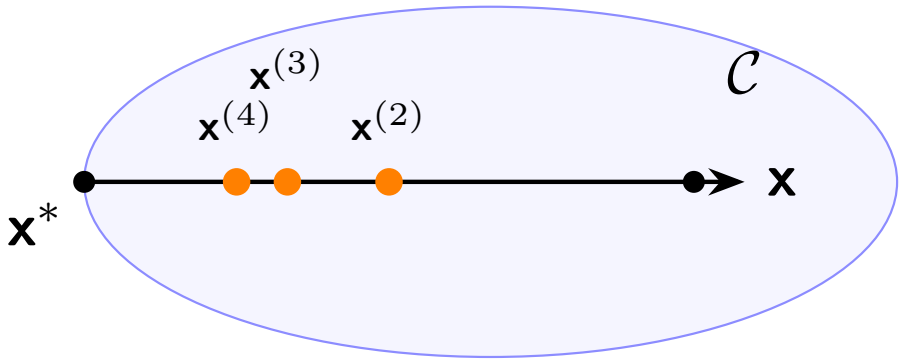
From the VI,

$$\nabla f(\mathbf{x}^*)^T(\mathbf{x}^{(k)} - \mathbf{x}^*) \geq 0 \quad \Longrightarrow \quad \nabla f(\mathbf{x}^*)^T\mathbf{d} \geq 0.$$

Under LICQ, KKT FONC $\Leftrightarrow \nabla f(\mathbf{x}^*)^{\mathsf{T}}\mathbf{d} \geq 0$ for any $\mathbf{d} \in T_{\mathcal{C}}(\mathbf{x}^*)$.
Farkas' alternative: exactly one of the following holds.

- $\nabla f(\mathbf{x}^*) + \nabla \mathbf{g}(\mathbf{x}^*)\boldsymbol{\lambda}^* + \nabla \mathbf{h}_J(\mathbf{x}^*)\boldsymbol{\nu}_J^* = \mathbf{0}$ for some $\boldsymbol{\lambda}^* \in \mathbf{R}^m$ and $\boldsymbol{\nu}_J^* \in \mathbf{R}_+^{|J|}$;
- there is $\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*)$ such that $\nabla f(\mathbf{x}^*)^{\mathsf{T}}\mathbf{d} < 0$.

KKT FONC ⇒ VI



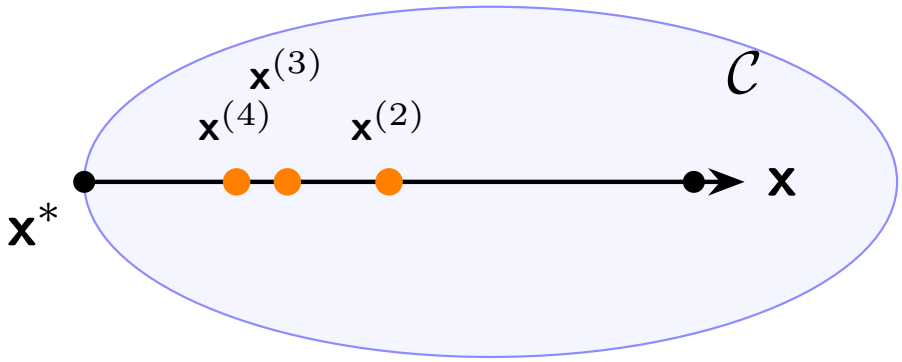
For any $\mathbf{x} \in \mathcal{C}$, define $\mathbf{d} = \mathbf{x} - \mathbf{x}^*$,

---

From $T_{\mathcal{C}}(\mathbf{x}^*) \subseteq T_{\text{lin}}(\mathbf{x}^*)$, KKT FONC $\Rightarrow \nabla f(\mathbf{x}^*)^{\mathsf{T}}\mathbf{d} \geq 0$ for any $\mathbf{d} \in T_{\mathcal{C}}(\mathbf{x}^*)$.
Farkas' alternative: exactly one of the following holds.

- $\nabla f(\mathbf{x}^*) + \nabla \mathbf{g}(\mathbf{x}^*)\boldsymbol{\lambda}^* + \nabla \mathbf{h}_J(\mathbf{x}^*)\boldsymbol{\nu}_J^* = \mathbf{0}$ for some $\boldsymbol{\lambda}^* \in \mathbf{R}^m$ and $\boldsymbol{\nu}_J^* \in \mathbf{R}_+^{|J|}$;
- there is $\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*)$ such that $\nabla f(\mathbf{x}^*)^{\mathsf{T}}\mathbf{d} < 0$.

KKT FONC ⇒ VI



For any $\mathbf{x} \in \mathcal{C}$, define $\mathbf{d} = \mathbf{x} - \mathbf{x}^*$,

$$\mathbf{x}^{(k)} = \mathbf{x}^* + \frac{\mathbf{d}}{k} \in \mathcal{C}, \quad \mathbf{x}^{(k)} \to \mathbf{x}^*, \quad t^{(k)} = \frac{1}{k}, \quad \frac{\mathbf{x}^{(k)} - \mathbf{x}^*}{t^{(k)}} = \mathbf{d}.$$

So $0 \leq \nabla f(\mathbf{x}^*)^T \mathbf{d} = \nabla f(\mathbf{x}^*)^T (\mathbf{x} - \mathbf{x}^*)$ (VI).

From $T_{\mathcal{C}}(\mathbf{x}^*) \subseteq T_{\text{lin}}(\mathbf{x}^*)$, KKT FONC $\Rightarrow \nabla f(\mathbf{x}^*)^{\mathsf{T}} \mathbf{d} \geq 0$ for any $\mathbf{d} \in T_{\mathcal{C}}(\mathbf{x}^*)$.
Farkas' alternative: exactly one of the following holds.

- $\nabla f(\mathbf{x}^*) + \nabla \mathbf{g}(\mathbf{x}^*) \boldsymbol{\lambda}^* + \nabla \mathbf{h}_J(\mathbf{x}^*) \boldsymbol{\nu}_J^* = \mathbf{0}$ for some $\boldsymbol{\lambda}^* \in \mathbf{R}^m$ and $\boldsymbol{\nu}_J^* \in \mathbf{R}_+^{|J|}$;
- there is $\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*)$ such that $\nabla f(\mathbf{x}^*)^{\mathsf{T}} \mathbf{d} < 0$.

Convex KKT: the precise theorem

- Global min $\Leftrightarrow$ VI.
- VI + LICQ $\Rightarrow$ KKT FONC.
- KKT FONC $\Rightarrow$ VI.

Theorem 14.3

Consider the canonical convex program and a feasible point $\mathbf{x}^*$.

1. *If* $\mathbf{x}^*$ *is a KKT point, it is a global minimizer.*
2. *If* $\mathbf{x}^*$ *is a global minimizer and LICQ holds at* $\mathbf{x}^*$, *it admits unique KKT multipliers.*

Consequently, under LICQ at $\mathbf{x}^*$,

$$\boxed{\mathbf{x}^* \text{ is a global minimizer} \quad \Longleftrightarrow \quad \mathbf{x}^* \text{ is a KKT point.}}$$

NB. Existence of a global minimizer needs a separate well-posedness check.

Trap 1: a convex minimizer may have no KKT certificate

$$\min_{x \in \mathbf{R}} x \qquad \text{s. t.} \qquad x^2 \leq 0.$$

- Feasible set $\mathcal{C} = \{0\}$: $x^* = 0$ is a global minimizer.
- Stationarity would require $1 + \nu \cdot 0 = 0$: impossible.
- LICQ fails: the active constraint gradient is zero.

$$T_{\mathcal{C}}(0) = \{0\}, \qquad T_{\text{lin}}(0) = \mathbf{R}.$$

LICQ at a minimizer guarantees the existence of multipliers.

Trap 2: KKT does not imply a unique global minimizer

$$\min_{(x_1,x_2)\in\mathbf{R}^2} x_1^2 \qquad \text{s. t.} \qquad x_2^2 - 1 \leq 0.$$

Every point in $\{(0, x_2) : -1 \leq x_2 \leq 1\}$ is a KKT point, with $\nu = 0$, and a global minimizer.

- LICQ holds at all of these points.
- Multipliers can be unique even when minimizers are not.

KKT FONC + (strictly convex f on $\mathcal{C}$ or KKT SOSC) $\Longrightarrow$ uniqueness.

Strictly convex $f \not\Rightarrow$ KKT SOSC

$$\min_{\mathbf{x}\in\mathbf{R}^2} x_1^4 + x_2^2 \qquad \text{s. t.} \qquad x_2 = 0.$$

- Strictly convex objective; $\mathbf{x}^* = \mathbf{0}$ is the unique minimizer.
- KKT multiplier: $\lambda^* = 0$.
- Critical cone: $C_{\text{crit}} = \{\mathbf{d} : d_2 = 0\}$.

$$\mathbf{H}_L(\mathbf{0}, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}, \qquad \mathbf{d}^\mathsf{T}\mathbf{H}_L\mathbf{d} = 0 \quad (\mathbf{d} \in C_{\text{crit}}).$$

Replacing strict convexity by C^2 **strong convexity** (L10) would work.

KKT SOSC $\not\Rightarrow$ strictly convex f

$$\min_{\mathbf{x}\in\mathbf{R}^2} x_2 \qquad \text{s. t.} \qquad h(\mathbf{x}) := x_1^2 - x_2 \le 0.$$

At $\mathbf{x}^* = \mathbf{0}$, stationarity gives $\nu^* = 1$.

$$C_{\text{crit}} = \{\mathbf{d} : d_2 = 0\}, \qquad \mathbf{H}_L(\mathbf{0}, 1) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

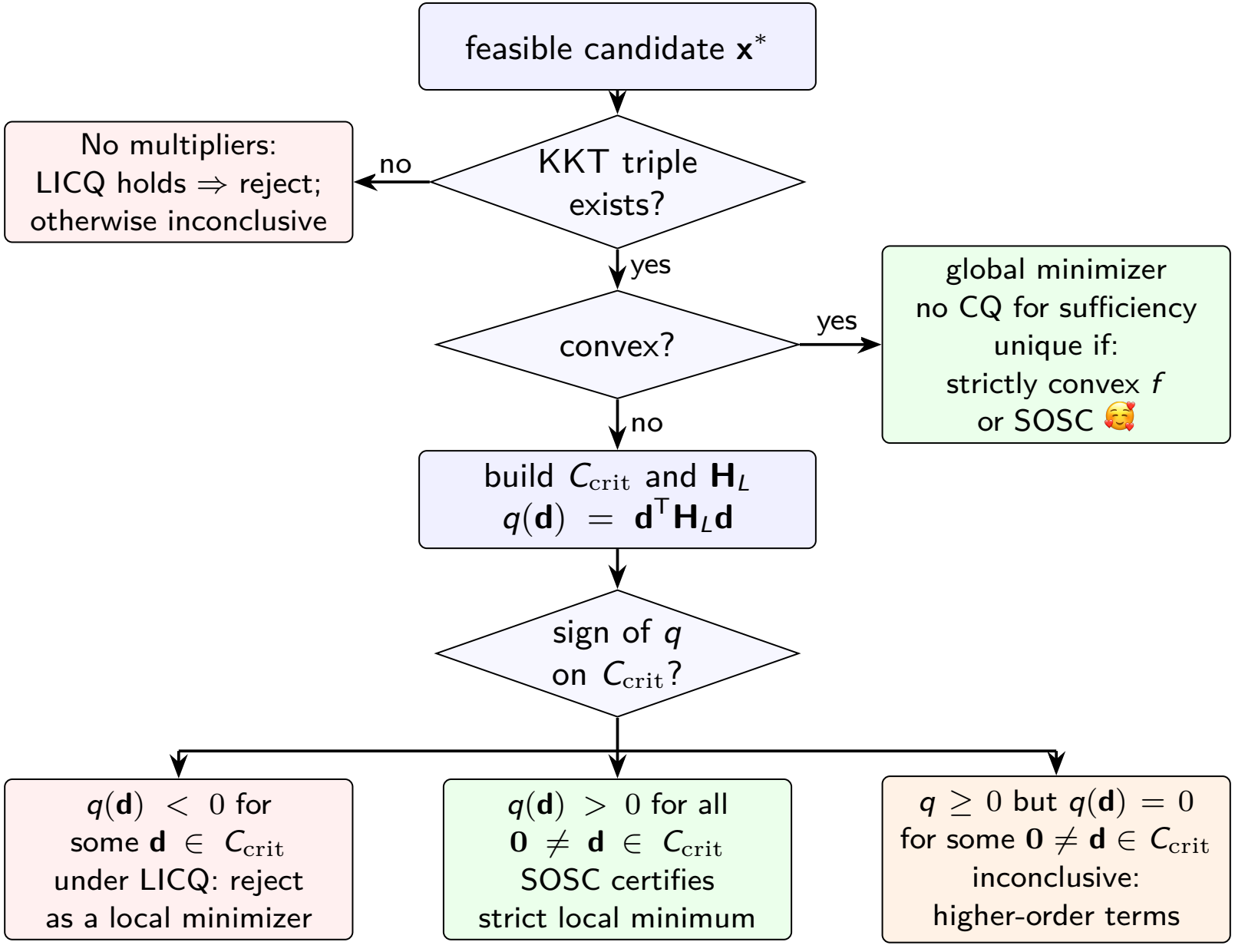
Thus

$$\mathbf{d}^{\mathsf{T}}\mathbf{H}_L\mathbf{d} = 2d_1^2 > 0 \qquad (\mathbf{0} \neq \mathbf{d} \in C_{\text{crit}}).$$

- Unique global minimizer: $x_2 \ge x_1^2 \ge 0$.
- Objective affine, not strictly convex.

Constraint curvature supplies the second-order certificate.

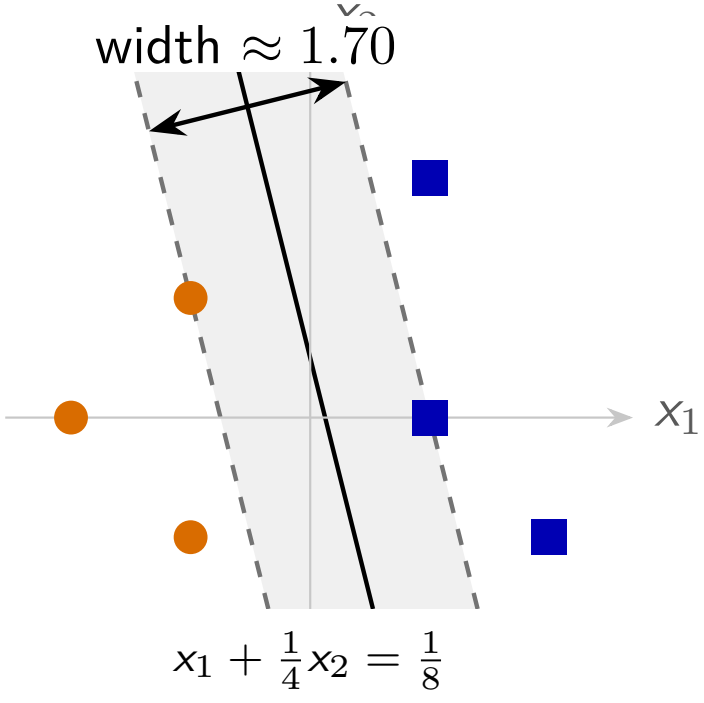
The complete KKT first-order and second-order workflow



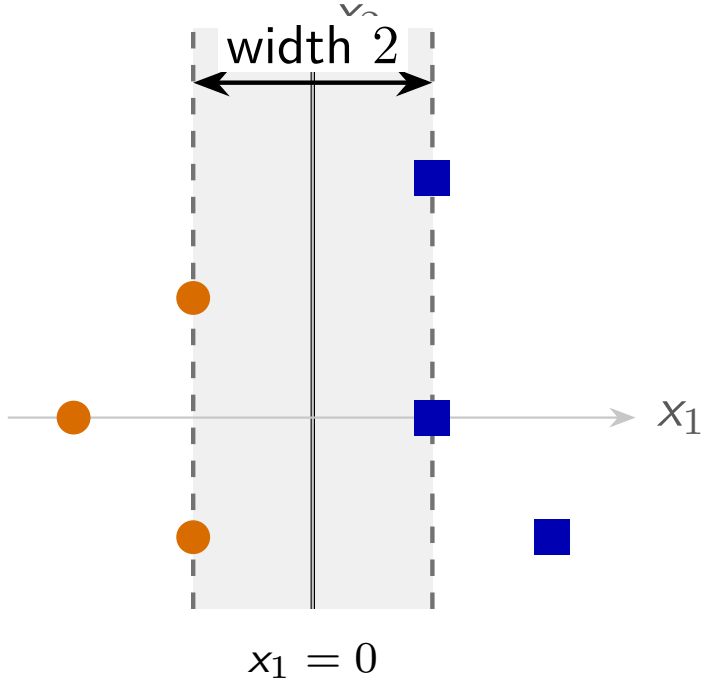
An application: a robust linear classifier

Linearly separable two classes of data: **blue** and **orange**.

Tilted separator



Maximum-margin separator



Support vector machine (SVM)

Among strict linear separators, **maximize the distance** between the two supporting hyperplanes.

Mathematical modeling of SVM

Finite data $(\mathbf{x}_i, y_i)$, $i = 1, \ldots, N$, with both classes $y_i \in \{-1, +1\}$ present.

- Decision hyperplane: $\mathbf{w}^\mathsf{T}\mathbf{x} + b = 0$, with $\mathbf{w} \neq \mathbf{0}$.
- After training: $\operatorname{sgn}(\mathbf{w}^\mathsf{T}\mathbf{x}_i + b) = y_i$
- $\alpha > 0$: geometric half-width of a centered empty band.

Mathematical modeling of SVM

Finite data $(\mathbf{x}_i, y_i)$, $i = 1, \ldots, N$, with both classes $y_i \in \{-1, +1\}$ present.

- Decision hyperplane: $\mathbf{w}^\mathsf{T}\mathbf{x} + b = 0$, with $\mathbf{w} \neq \mathbf{0}$.
- After training: $\operatorname{sgn}(\mathbf{w}^\mathsf{T}\mathbf{x}_i + b) = y_i$
- $\alpha > 0$: geometric half-width of a centered empty band.

The signed distance of observation i to the decision plane is

$$\frac{y_i(\mathbf{w}^\mathsf{T}\mathbf{x}_i + b)}{\|\mathbf{w}\|}.$$

Hence the geometric model is

$$\begin{aligned} \max_{\alpha>0,\, \mathbf{w}\neq\mathbf{0},\, b} \quad & \alpha \\ \text{s. t.} \quad & y_i(\mathbf{w}^\mathsf{T}\mathbf{x}_i + b) \geq \alpha\|\mathbf{w}\|, \quad i = 1, \ldots, N. \end{aligned}$$

Mathematical modeling of SVM (cont'd)

$$\max_{\alpha>0,\, \mathbf{w}\neq\mathbf{0},\, b} \quad \alpha, \quad \text{s. t.} \quad y_i(\mathbf{w}^{\mathsf{T}}\mathbf{x}_i + b) \geq \alpha\|\mathbf{w}\|, \quad i = 1, \ldots, N.$$

Set $\beta := \alpha\|\mathbf{w}\| > 0$. The same geometric problem becomes

$$\max_{\beta>0,\, \mathbf{w}\neq\mathbf{0},\, b} \quad \frac{\beta}{\|\mathbf{w}\|}, \quad \text{s. t.} \quad y_i(\mathbf{w}^{\mathsf{T}}\mathbf{x}_i + b) \geq \beta, \quad i = 1, \ldots, N.$$

Mathematical modeling of SVM (cont'd)

$$\max_{\alpha>0,\, \mathbf{w}\neq\mathbf{0},\, b} \quad \alpha, \quad \text{s. t.} \quad y_i(\mathbf{w}^{\mathsf{T}}\mathbf{x}_i + b) \geq \alpha\|\mathbf{w}\|, \quad i = 1, \ldots, N.$$

Set $\beta := \alpha\|\mathbf{w}\| > 0$. The same geometric problem becomes

$$\max_{\beta>0,\, \mathbf{w}\neq\mathbf{0},\, b} \quad \frac{\beta}{\|\mathbf{w}\|}, \quad \text{s. t.} \quad y_i(\mathbf{w}^{\mathsf{T}}\mathbf{x}_i + b) \geq \beta, \quad i = 1, \ldots, N.$$

Remove the arbitrary positive scale:

$$\widetilde{\mathbf{w}} := \frac{\mathbf{w}}{\beta}, \qquad \widetilde{b} := \frac{b}{\beta}, \qquad \alpha = \frac{1}{\|\widetilde{\mathbf{w}}\|}, \qquad y_i(\widetilde{\mathbf{w}}^{\mathsf{T}}\mathbf{x}_i + \widetilde{b}) \geq 1.$$

Maximizing $1/\|\widetilde{\mathbf{w}}\|$ is equivalent to minimizing $\frac{1}{2}\|\widetilde{\mathbf{w}}\|^2$. No $\mathbf{w} \neq \mathbf{0}$!

$$\min_{\mathbf{w},b} \ \frac{1}{2}\|\mathbf{w}\|^2, \quad \text{s. t.} \quad y_i(\mathbf{w}^{\mathsf{T}}\mathbf{x}_i + b) \geq 1, \quad i = 1, \ldots, N.$$

SVM is a canonical convex program

$$\begin{array}{ll} \min\limits_{\mathbf{w},b} & \frac{1}{2}\|\mathbf{w}\|^2 \\ \text{s. t.} & 1 - y_i(\mathbf{w}^\mathsf{T}\mathbf{x}_i + b) \leq 0, \quad i = 1, \ldots, N. \end{array}$$

- Convex quadratic objective; affine inequalities.
- Margin planes: $\mathbf{w}^\mathsf{T}\mathbf{x} + b = \pm 1$, with full band width $2/\|\mathbf{w}\|$.
- Strict separability $\Rightarrow$ feasibility and attainment.
- Every KKT point is globally optimal.
- The objective is not strictly convex in $(\mathbf{w}, b)$ jointly.

NB. Affine constraints ensure $T_{\text{lin}} = T_{\mathcal{C}}$ without LICQ.

The SVM Lagrangian and stationarity

$$L(\mathbf{w}, b, \boldsymbol{\nu}) = \frac{1}{2}\|\mathbf{w}\|^2 + \sum_{i=1}^{N} \nu_i [1 - y_i(\mathbf{w}^\mathsf{T}\mathbf{x}_i + b)].$$

$$\nabla_{\mathbf{w}} L = \mathbf{0} \quad \Longrightarrow \quad \boxed{\mathbf{w}^* = \sum_i \nu_i^* y_i \mathbf{x}_i},$$

$$\partial_b L = 0 \quad \Longrightarrow \quad \boxed{\sum_i \nu_i^* y_i = 0}.$$

- Only observations (from both classes) with $\nu_i^* > 0$ contribute to $\mathbf{w}^*$.
- Total multiplier weight balances between the classes.

The complete KKT system

$$\begin{cases} \mathbf{w}^* = \sum_i \nu_i^* y_i \mathbf{x}_i, \\ \sum_i \nu_i^* y_i = 0, \\ y_i(\mathbf{w}^{*\mathsf{T}} \mathbf{x}_i + b^*) \geq 1, \\ \nu_i^* \geq 0, \\ \nu_i^*[1 - y_i(\mathbf{w}^{*\mathsf{T}} \mathbf{x}_i + b^*)] = 0. \end{cases}$$

Every solution of this KKT system solves SVM globally.

Support vectors: touching versus loaded

Use a geometric convention for SVM:

$$\text{support vector} \iff y_i(\mathbf{w}^{*\mathsf{T}}\mathbf{x}_i + b^*) = 1 \iff \text{active constraint}.$$

Complementarity gives only the one-way implication

$$\nu_i^* > 0 \implies \text{support vector}.$$

- Loaded support vector: $\nu_i^* > 0$.
- **A support vector may merely touch**: $\nu_i^* = 0$.
- **Inactive observation**: necessarily $\nu_i^* = 0$.

Recover the intercept from any support vector

For any active observation, even if $\nu_i^* = 0$,

$$y_i(\mathbf{w}^{*\mathsf{T}}\mathbf{x}_i + b^*) = 1 \quad \Longrightarrow \quad \boxed{b^* = y_i - \mathbf{w}^{*\mathsf{T}}\mathbf{x}_i}.$$

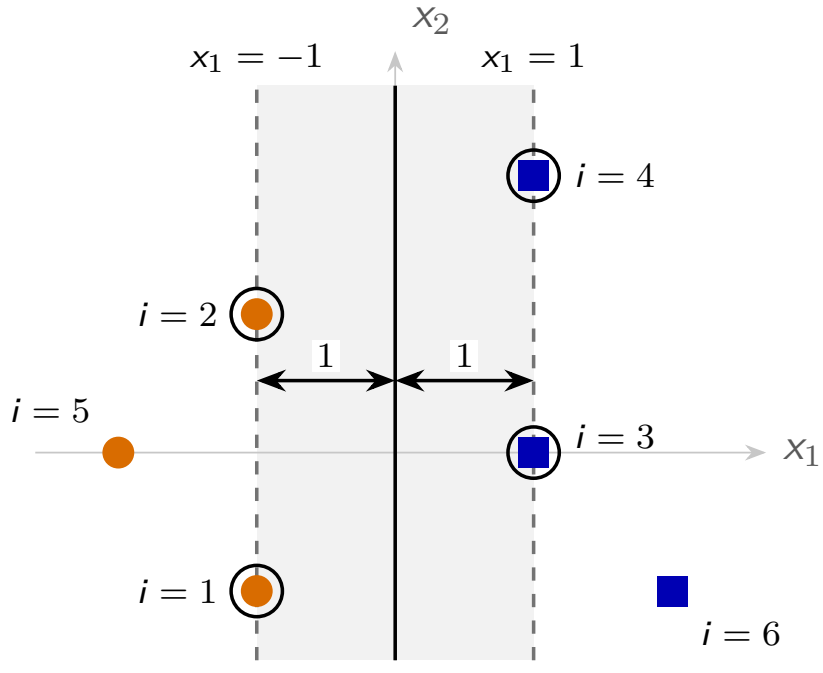
- At an exact solution, every active point gives the same b^*.
- Numerically, let $A \neq \varnothing$ index reliable near-active observations and average:

$$b \approx \frac{1}{|A|} \sum_{i \in A} (y_i - \mathbf{w}^{\mathsf{T}}\mathbf{x}_i).$$

- Both-class SVM has unique $(\mathbf{w}^*, b^*)$, even if multipliers are nonunique.

A worked example: six observations

i	1	2	3	4	5	6
$x_{i,1}$	-1	-1	1	1	-2	2
$x_{i,2}$	-1	1	0	2	0	-1
y_i	-1	-1	$+1$	$+1$	-1	$+1$



rings: active constraints

- The data are strictly separable.
- Points $1, 2$ and $3, 4$ suggest supporting lines $x_1 = -1, +1$.
- Candidate: $\mathbf{w}^* = (1, 0)^\mathsf{T}$, $b^* = 0$.

Solve the example's multiplier equations

At $(\mathbf{w}^*, b^*) = ((1,0)^\mathsf{T}, 0)$, points 5 and 6 are inactive observations. Thus $\nu_5 = \nu_6 = 0$; stationarity becomes

$$\begin{cases} \mathbf{w}^* = \sum_i \nu_i^* y_i \mathbf{x}_i, \\ \sum_i \nu_i^* y_i = 0 \end{cases} \quad \Rightarrow \quad \begin{cases} \nu_1 + \nu_2 + \nu_3 + \nu_4 = 1, \\ \nu_1 - \nu_2 + 2\nu_4 = 0, \\ -\nu_1 - \nu_2 + \nu_3 + \nu_4 = 0. \end{cases}$$

Solve the example's multiplier equations

At $(\mathbf{w}^*, b^*) = ((1,0)^\mathsf{T}, 0)$, points 5 and 6 are inactive observations. Thus $\nu_5 = \nu_6 = 0$; stationarity becomes

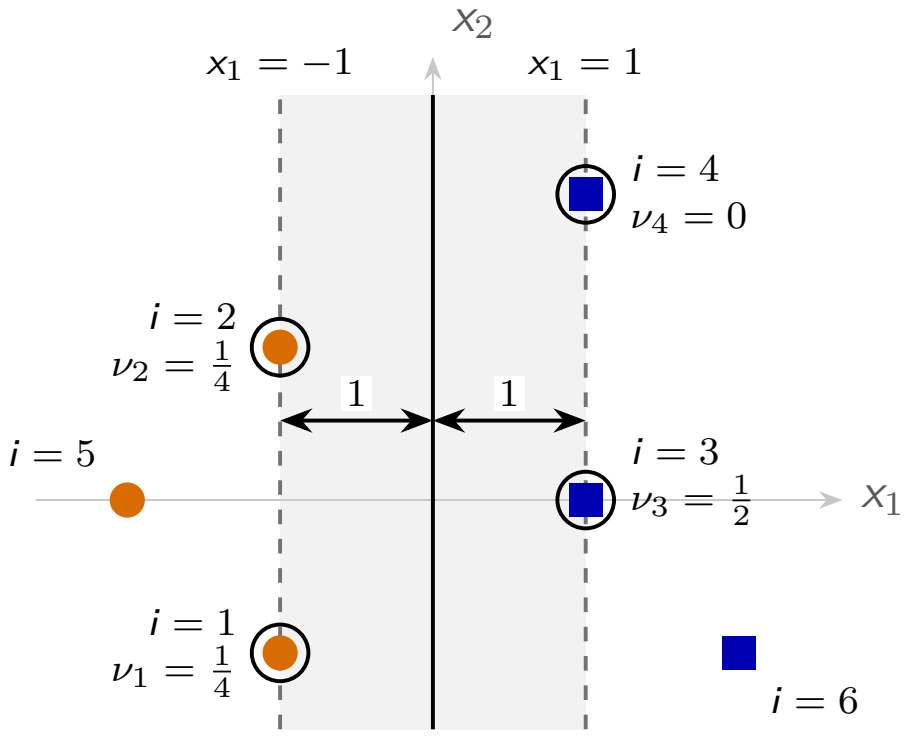
$$\begin{cases} \mathbf{w}^* = \sum_i \nu_i^* y_i \mathbf{x}_i, \\ \sum_i \nu_i^* y_i = 0 \end{cases} \quad \Rightarrow \quad \begin{cases} \nu_1 + \nu_2 + \nu_3 + \nu_4 = 1, \\ \nu_1 - \nu_2 + 2\nu_4 = 0, \\ -\nu_1 - \nu_2 + \nu_3 + \nu_4 = 0. \end{cases}$$

One nonnegative solution is

$$\boxed{\boldsymbol{\nu}^* = (1/4, 1/4, 1/2, 0, 0, 0)^\mathsf{T}}.$$

Feasibility and complementarity hold. Hence global optimum: $f^* = 1/2$, margin width 2.

The solution: four contacts, three loads

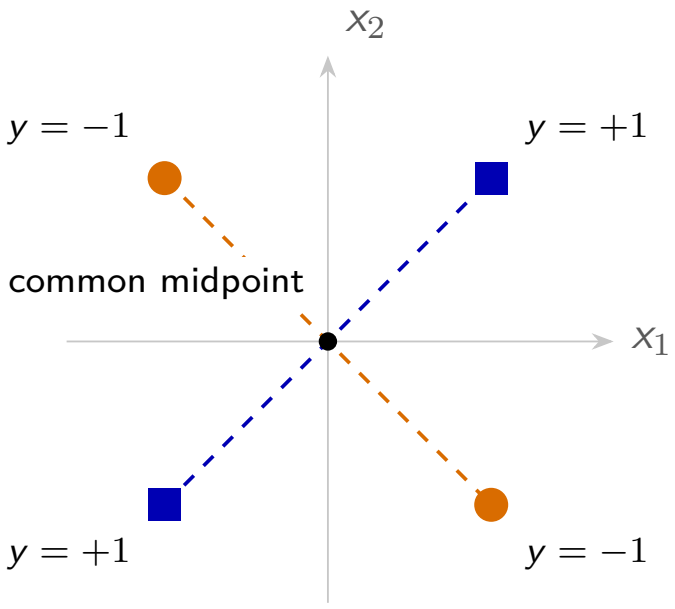


rings: active constraints

$i = 1, 2, 3$: loaded $\quad$ $i = 4$: active, unloaded

- Geometric support vectors: $1, 2, 3, 4$.
- Loaded for this certificate: $1, 2, 3$.
- Point 4: active, but $\nu_4 = 0$.
- Contact need not carry load.

When no strict linear separator exists

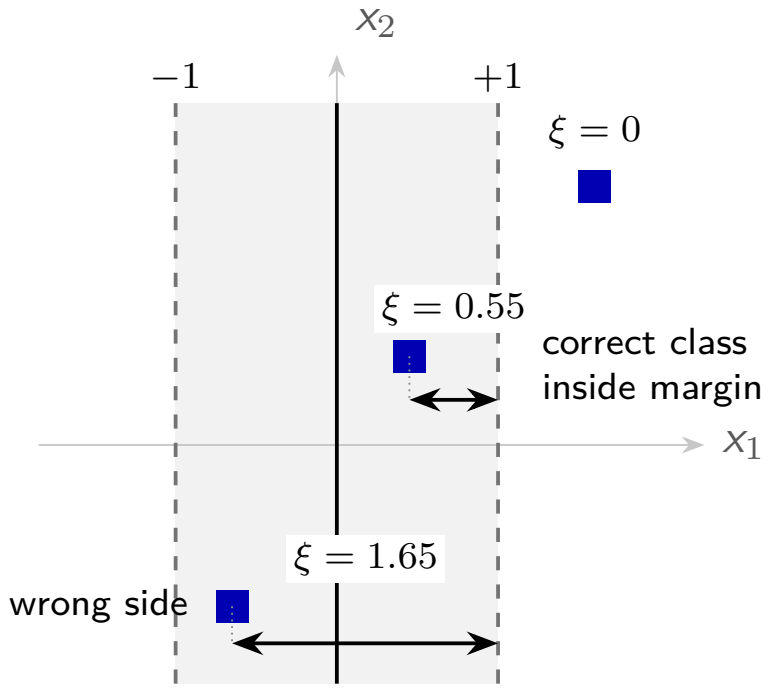


- Opposing class convex hulls meet: no strict separator.
- Hard-margin constraints are infeasible.

Question

Can we allow violations, but keep a convex optimization problem?

Introducing slack variables



fixed separator $x_1 = 0$ all displayed points: $y_i = +1$

For a candidate, write $s_i = \mathbf{w}^\mathsf{T}\mathbf{x}_i + b$ and choose the smallest feasible slack:

$$\xi_i = \max\{0, 1 - y_i s_i\}.$$

- $\xi_i = 0$: on or beyond the correct margin.
- $0 < \xi_i < 1$: correct side, inside margin.
- $\xi_i = 1$: on the separator.
- $\xi_i > 1$: wrong side.

Relaxed constraints: $y_i(\mathbf{w}^\mathsf{T}\mathbf{x}_i + b) + \xi_i \geq 1,\ \xi_i \geq 0.$

Soft-margin SVM

Introduce $\xi_i \geq 0$ and a penalty parameter $C > 0$:

$$\begin{aligned} \min_{\mathbf{w},b,\boldsymbol{\xi}} \quad & \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{N}\xi_i \\ \text{s. t.} \quad & 1 - y_i(\mathbf{w}^\mathsf{T}\mathbf{x}_i + b) - \xi_i \leq 0, \\ & -\xi_i \leq 0, \qquad i = 1, \ldots, N. \end{aligned}$$

- Convex quadratic-plus-linear objective; affine constraints.
- Always feasible; the finite-data problem attains a minimum.
- Larger C: margin violations are more expensive.

Take-home messages

- Canonical convex programs: KKT FONC $\Rightarrow$ global min.
- Under LICQ at a feasible point, global min $\Longleftrightarrow$ KKT.
- Soft-margin and hard-margin SVMs: convex quadratic programs.
- Next: (primal) numerical methods for constrained problems.

Appendix: why the margin formula is exact

Let $H_c = \{\mathbf{x} : \mathbf{w}^\mathsf{T}\mathbf{x} + b = c\}$, $\mathbf{w} \neq \mathbf{0}$. For $\mathbf{x} \in H_{c_1}$ and $\mathbf{y} \in H_{c_2}$,

$$|c_2 - c_1| = |\mathbf{w}^\mathsf{T}(\mathbf{y} - \mathbf{x})| \leq \|\mathbf{w}\| \, \|\mathbf{y} - \mathbf{x}\|.$$

The bound is attained by

$$\mathbf{y} = \mathbf{x} + \frac{c_2 - c_1}{\|\mathbf{w}\|^2}\mathbf{w}.$$

Hence

$$\operatorname{dist}(H_{c_1}, H_{c_2}) = \frac{|c_2 - c_1|}{\|\mathbf{w}\|}.$$

Set $c_1 = -1, c_2 = 1$ to get the full margin width $2/\|\mathbf{w}\|$.